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Question

Properties of TRAPPIST-1 Exoplanets Compared to Earth's Properties

Planet	Orbital period (days)	Planet radius (Earth radii)
Earth	365.26	1
TRAPPIST-1b	1.51	1.09
TRAPPIST-1d	4.05	0.77
TRAPPIST-1e	6.1	0.92
TRAPPIST-1f	9.21	1.04

TRAPPIST-1 is a planetary system with a central red dwarf star and several Earth-sized exoplanets. Each planet in this system is so close in orbit to the star that the gravitational forces acting between the planet and the star cause the planet's rotational period around its axis and orbital period around the star to be equal. For example, the orbital period of TRAPPIST-1e is ______

Which choice most effectively uses data from the table to complete the example?

Answer

A. 1.51 days, which is the shortest rotational period of any of the TRAPPIST planets.

B. 6.1 days, and its rotational period is also 6.1 days.

C. 9.21 days, which is the same as the rotational period of TRAPPIST-1f.

D. 6.1 days, and its rotational period is 0.92 days.